Asset Pricing and Equity Valuation Analysis of ASML Holding N.V.

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EQUITY RESEARCH | INITIATING COVERAGE

ASML HOLDING N.V.

ASML NA | Euronext Amsterdam | Semiconductor Capital Equipment (WFE)

HOLD | PT: €578 | Current: €1,150 | Downside: (49.8%)

DCF Value	Comps Value	Mkt Cap	WACC	Alpha	Downside (DCF)
€551	**€604**	**€440bn**	**9.0%**	**−31.1%**	**(52.1%)**
Intrinsic/Share	Peer Avg	Mar-2026	CAPM-Derived	Implied−Reqd	vs €1,150

The Irreplaceable Enabler - Priced for Perfection

Base-case DCF and peer comps imply significant overvaluation relative to a €1,150 market price. Strong fundamentals support Hold.

1. EXECUTIVE SUMMARY & INVESTMENT HIGHLIGHTS

ASML Holding N.V. is the sole manufacturer of EUV lithography systems - a position of the structural monopoly which is unmatched in the semiconductor supply chain. Our base-case FCFF DCF yields an intrinsic value of €550.96/share and comparable company analysis implies €604.40/share, against a market price of €1,150. We initiate with a HOLD. The business is exceptional; the price is not. CAPM analysis (Rf 2.8%, β 1.35, ERP 5.5%) requires a 10.2% return; the model-implied total return is −20.9%, which is generating a negative alpha of −31.1% - confirming that the current price more than fully prices ASML's structural advantages under base-case assumptions.

Table 1.1 Executive Summary Table

Highlight	Key Analytical Point
Structural Monopoly	Sole EUV supplier; 10–15 year lead over nearest rival SMEE. 4,000-supplier ecosystem is unreplicable within any commercial timeframe. Monopoly justifies premium - but not at 68% above peers.
Valuation Gap	DCF €551 and Comps €604 converge - two independent methods confirm ~50% overvaluation. Gap reflects market pricing AI option value and High-NA ramp beyond our conservative base case.
Negative Alpha −31.1%	Implied return −20.9% vs. required 10.2%. At €1,150, investors are not compensated for ASML's systematic (WFE cyclical) risk. Sell-side consensus targets €955–€1,700 (median ~€1,400), reflecting more aggressive growth assumptions.

First-Class Fundamentals	FY2025: Revenue €32,667m (+15.6%), EBIT margin 34.6% (5Y high), FCF margin 33.8%, FCF/NI 114.8%, net cash €11bn. EPS CAGR 16.3% over FY2021–2025. Buyback of €5.95bn in FY2025.
China Export Risk	Dutch export controls progressively restrict DUV/EUV to China (historically ~30% of revenue). Full embargo could reduce FY2026E revenue by €4–6bn, depressing intrinsic value to ~€480–500/share.

2. COMPANY OVERVIEW

ASML designs and manufactures photolithography systems spanning DUV (ArF/KrF), EUV (NXE:3600/3800, 13.5nm), and High-NA EUV (EXE:5000, 0.55NA, ~€350m/unit). Founded in the 1984 as a Philips-ASM JV, it has evolved into the most strategically critical supplier in the feild semiconductor value chain. TSMC, Samsung, and Intel have no alternative for sub-5nm manufacturing. The EUV system requires 100,000+ components from ~4,000 specialised suppliers - a supply chain architecture that constitutes a structural entry barrier of the highest order.

Revenue is driven by:

(1) EUV system sales to leading-edge chipmakers;

(2) DUV systems for mature nodes;

(3) Installed Base Management (IBM) - service, upgrades, and refurbishment of the 7,000+ unit fleet - growing at ~8% CAGR and providing margin-accretive recurring revenue.

The AI infrastructure supercycle creates a demand multiplier: AI capex flows from hyperscalers to chip designers to TSMC/Samsung and ultimately to ASML, the sole supplier of the enabling manufacturing technology. High-NA EUV enters volume production in FY2026–2028, representing a per-unit revenue step-change and a new source of operating leverage.

3. OPERATIONAL PERFORMANCE ANALYSIS

Revenue grew from €18,611m (FY2021) to €32,667m (FY2025) at a 15.1% four-year CAGR

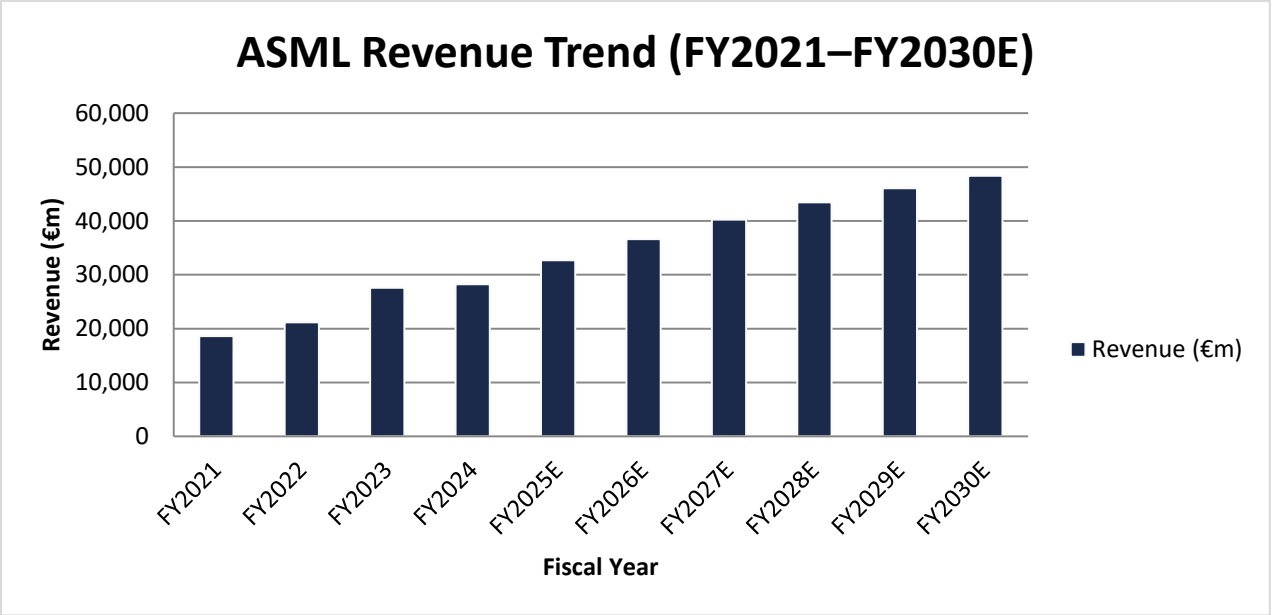


Figure 1: ASML Revenue Trend (FY2021–FY2030E)

the trajectory was not a line it was more like a wave. This wave followed the ups and downs of the semiconductor equipment business. The 30.2% surge in FY2023 was followed by near-stagnation in FY2024 (+2.5%), This happened because companies were reducing their stock after COVID and they were also dealing with the news about Dutch export controls. Order intake collapsed 52% from FY2023 (€39,400m) to FY2024 (€18,900m), This big drop told us that the business was at a point and it also led to the stock price going down to 542 euros in April 2025. The recovery in FY2025 orders to €27,200m and record revenue/margin confirms a cyclical - not structural - deterioration.

Table 3.1 Operational Performance Analysis

Metric (€m)	**FY2021A**	**FY2022A**	**FY2023A**	**FY2024A**	**FY2025A**	**5Y CAGR**
Revenue	18,611	21,173	27,559	28,261	**32,667**	**15.1%**
YoY Growth	-	+13.8%	+30.2%	+2.5%	+15.6%	-
EBIT Margin	30.8%	28.1%	32.7%	31.9%	**34.6%**	-
EPS Diluted (€)	€13.49	€13.61	€21.00	€19.24	**€24.71**	**16.3%**
Order Intake	26,000	40,400	39,400	**18,900**	27,200	-
Order Backlog	33,000	40,400	39,400	36,000	**38,800**	-

4. FINANCIAL PERFORMANCE ANALYSIS

ASML's profitability traces a V-shaped recovery:

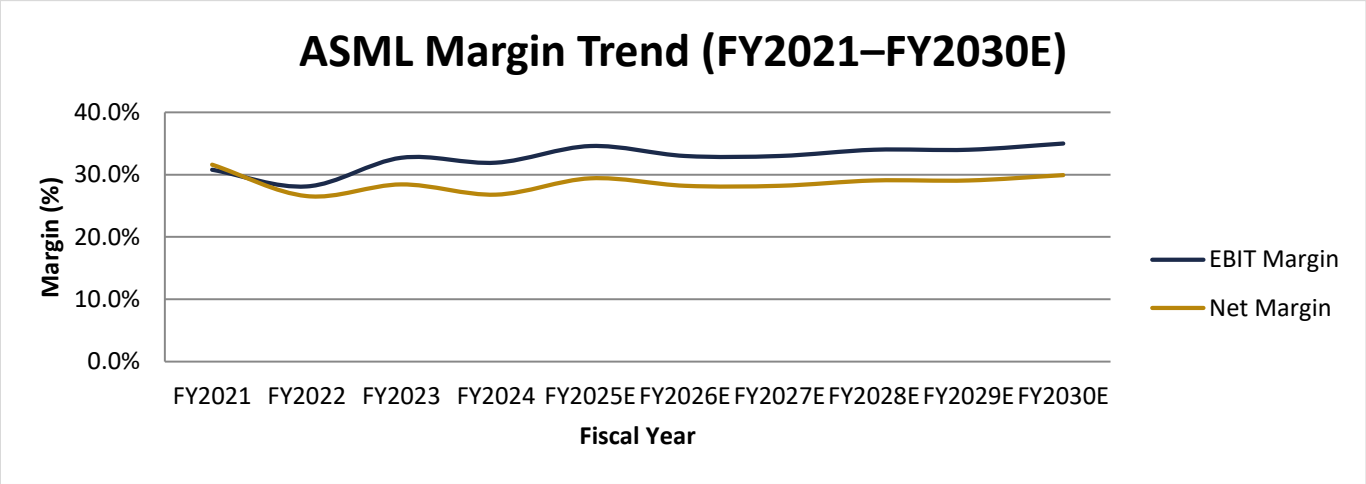


Figure 2: EBIT and Net Margin Trend (FY2021–FY2030E)

The EBIT margin of ASML went down from 30.8 percent in FY2021 to 28.1 percent in FY2022 because the company spent a lot on High-NA research and development.. Then the EBIT margin of ASML went up to 34.6 percent in FY2025, which is the highest it has been in five years. This shows that ASML is making money as it sells more.

The net margin of ASML has been over the place going from 26.6 percent to 31.6 percent. This is because the products ASML sells are different and that affects the money the company makes. When ASML sells a lot of EUV products it makes a lot of money.. When it sells a lot of DUV products it does not make as much. ASML made a lot of cash in FY2025 with free cash flow being 114.8 percent of income. The company had €11,028 million in cash flow. This is because customers pay ahead of time for EUV products, which takes a time to make.

At the end of FY2025 ASML had €11,000 million in cash. It bought back €5.95 billion of its own shares for about €800 per share. This shows that the people in charge of ASML are okay, with the share price being lower than it is today which's €1,150.

Table 4.1 Financial Performance Analysis

Metric	FY2021A	FY2022A	FY2023A	FY2024A	FY2025A	Trend
Gross Margin	52.7%	49.9%	51.3%	51.0%	**52.8%**	↑ Recovering
EBIT Margin	30.8%	28.1%	32.7%	31.9%	**34.6%**	↑ **5Y High**
Net Margin	31.6%	26.6%	28.4%	26.8%	29.4%	↑ Improving
FCF Margin	24.2%	24.6%	25.0%	32.1%	**33.8%**	↑ **Best-in-Class**

Net Income (€m)	5,882	5,624	7,837	7,572	**9,609**	CAGR 13.1%
P/E Trailing (×)	48.2×	38.2×	28.6×	34.3×	29.8×	**NTM: 40×**

The earnings per share went up from 19.24 euros in the year 2024 to 24.71 euros in the financial year 2025. That is an increase of 28.4 percent. The company bought back some of its shares which helped to reduce the total number of shares. This means there were 420 million shares before and now there are 383 million shares.

The company is also doing a job of controlling how much money it spends on new things. It spent 1,631 million euros on things in the financial year 2025. This is 5 percent of the money the company made that year. The company had spent money in the financial year 2024 when it was building a lot of new things.

The tax rules in the Netherlands are also helping the company. The company only has to pay 14.5 percent of its earnings in taxes. This is an advantage for the company and it helps the company to make more money. The earnings per share of the company is getting a boost from this tax rule. The company is benefiting from the Dutch innovation box tax regime and the earnings, per share is increasing because of it.

5. FORECAST ASSUMPTIONS

The growth in revenue is estimated to be 12.0% in FY2026 - based on the order backlog of 38,800m, management target in the midpoint of 32.5bn and agreed range of 35-38bn. High-NA EUV is adopting on an S-curve, and will decrease by 12 to 5 percent by FY2030. EBIT margin would rise to 33% in FY202627 (minor dilution due to cost of High-NA ramp), and 35% by FY2030 as IBM scales and High-NA yields become fully productive - in line with the own 2030 financial forecast of ASML itself.

Table 5.1 Forecast Assumptions

Assumption	FY26E	FY27E	FY28E	FY29E	FY30E	Rationale
Revenue Growth	12.0%	10.0%	8.0%	6.0%	5.0%	High-NA ramp + AI capex; moderation as cycle matures; aligns to ASML 2030 scenario midpoint
Revenue (€m)	36,587	40,246	43,465	46,073	48,377	5Y CAGR ~8.2% from FY2025A base of €32,667m
EBIT Margin	33.0%	33.0%	34.0%	34.0%	35.0%	Slight near-term dilution (High-NA ramp); IBM-driven recovery; GM target 54–56% by FY2030
Tax / D&A / Capex	14.5%	3.0%	5.0%	0.5%	Stable	Dutch innovation box (14.5% tax); D&A 3% per FY2025 actual €1,026m;

						Capex 5% normalised steady-state; ΔWC 0.5%

6. VALUATION ANALYSIS

6a. DCF Valuation - FCFF Method

We discount 5 years of explicit FCFF projections (FY2026-2030) and Gordon Growth terminal value of WACC 9.0. FCFF is applied as the leverage of ASML is close to zero (net cash €11bn) and the values of firm and equity are close to each other. The terminal value of TGR 3.0% is 77.3 percent of EV - high but suitable to a compounding monopoly with structural 10-20 year technology runway. Accuracy is taken care of through mid-year discounting since March 2026 (periods 0.5y-4.5y)..

Table 6.1: DCF Forecast and Free Cash Flow Projection (FY2026E–FY2030E)

(€m)	FY2026E	FY2027E	FY2028E	FY2029E	FY2030E	TV / Bridge
Revenue	36,587	40,246	43,465	46,073	48,377	-
EBIT Margin	33.0%	33.0%	34.0%	34.0%	35.0%	-
NOPAT	10,323	11,355	12,635	13,394	14,477	-
FCFF	**9,408**	**10,349**	**11,549**	**12,242**	**13,267**	**TV: €227,757m**
Discount Factor	0.9578	0.8787	0.8062	0.7396	0.6785	PV(TV): 0.6785
PV of FCFF	**9,012**	**9,094**	**9,310**	**9,054**	**9,003**	**PV(TV): €154,544m**

Sum of PV FCFFs: €45,473m **PV of Terminal Value: €154,544m (77.3% of EV)** **Enterprise Value: €200,017m** (+) Net Cash: €11,000m → Equity Value: €211,017m	**Intrinsic Value per Share:** **€550.96** vs. Market Price: €1,150 **DCF Downside: (52.1%)**

The output is verified to be sound in plausible WACC/TGR values through sensitivity analysis. The model only converges to the current pricing with optimistic assumptions (WACC 8.0%, TGR 4.0%) which need assumptions that are materially higher than base case.

Table 6.2: DCF Valuation Sensitivity Analysis (WACC vs Terminal Growth Rate

WACC \ TGR	2.0%	2.5%	3.0% ◄ Base	3.5%	4.0%
8.0%	€664	€720	€787	€870	€975
8.5%	€603	€649	€703	€769	€851
9.0% ◄	€550	€590	**€635**	€689	€755

9.5%	€505	€539	€577	€622	€677
10.0%	€466	€495	€528	€567	€614

Advantages: FCFF is not related to capital structure; long-term cash flows are captured; in theory, it is ideal in monopolistic firms. Limitations: 77% terminal value weight generates WACC sensitivity; is unable to sufficiently model the High-NA EUV option value; Gordon Growth assumes steady-state by FY2030 which can understate the structural runway.

6b. Comparable Company Valuation

Table 6.3 Trading Comparables — Peer Multiples and Relative Positioning

Company	Ticker	Mkt Cap	NTM P/E	EV/EBITDA	Rev Gr.	Gross Margin
ASML Holding	ASML NA	€440bn	**40.0×**	**31.0×**	12.0%	**52.8%**
Applied Materials	AMAT US	$120bn	20.0×	15.0×	8.0%	47.0%
Lam Research	LRCX US	$80bn	22.0×	16.0×	9.0%	48.0%
KLA Corporation	KLAC US	$70bn	25.0×	18.0×	7.0%	60.0%
Tokyo Electron	8035 JP	€85bn	28.0×	20.0×	10.0%	46.0%
Peer Average (ex-ASML)	-	€89bn	**23.75×**	**17.25×**	**8.5%**	**50.3%**
ASML Premium	-	-	**+68.4%**	**+79.7%**	+3.5pp	**P/E Implied: €587 | EV/EBITDA: €622 | Avg: €604**

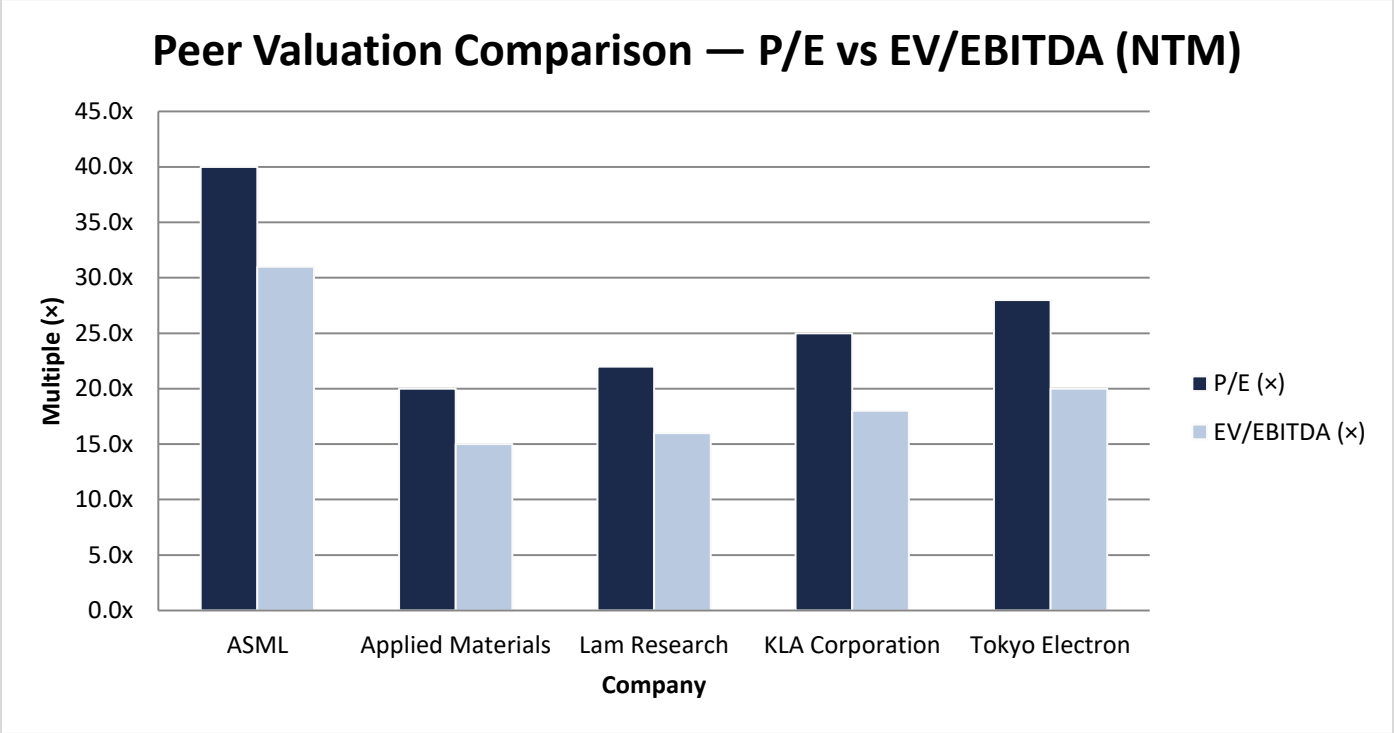


Figure 3: Peer Valuation Comparison — P/E vs EV/EBITDA

Using peer-average multiples on the NTM metrics at ASML results in the P/E-implied price of €587 (23.75× × €24.71 EPS) and the EV/EBITDA-implied price of €622 (17.25× × €13,171m EBITDA), which averages to 604 - validating the D The high P/E premium (68%) is partly justified by the monopoly status, pricing strength, and high gross margins (52.8% vs. the industry average 50.3%). But it is not inexhaustible - even with a justified premium of 4050% above peers would mean a price of 830890, significantly less than 1150. Peer group restriction is identified. No competitor has control and ASML is unique.The extra value should be based on the DCF minimum, not on comparisons.The price should be set by DCF floor not by multiples.

7. CAPM & ALPHA ANALYSIS

Table 7.1: CAPM Analysis - Required Return and Alpha Assessment

CAPM Parameter	Value	Formula/Source	Interpretation
Risk-Free Rate (Rf)	**2.80%**	10Y German Bund, Mar-2026	Post-ECB normalisation; slight rise from Apr-25 (2.5%)
Equity Beta (β)	**1.35**	5Y monthly, WFE sector	Above market - semiconductor equipment cyclicality; WFE spending swings ±50%+ through cycle
Equity Risk Premium (ERP)	**5.50%**	Damodaran W. Europe 2025	Western European implied ERP; standard academic benchmark

Required Return E(Ri) = Rf + β×ERP	**10.225%**	2.8%+(1.35×5.5%)	Minimum annual return to compensate for ASML's systematic risk - the investment hurdle rate
Implied Total Return (3Y convergence + yield)	**−20.85%**	(€551/€1,150)^(1/3)−1 + 0.94%	Model-implied return proxy: 3Y price convergence to intrinsic value + dividend yield 0.94% (€10.80/share ÷ €1,150)
ALPHA (Implied − Required)	**−31.1%**	Implied minus CAPM	Deeply negative: current price does not compensate investors for systematic risk under base-case assumptions

The negative alpha has to be interpreted with accuracy: the negative alpha does not imply that ASML will collapse, it does not imply that it is a mismanaged company. It implies, technically, that at the implied return of 1,150, the stock is not providing investors with the risk-adjusted returns that they are expected to receive on an annual 3-year convergence basis - i.e. the stock is failing to reward investors at the risk they assume according to our base-case assumptions. The empirical basis of the beta of 1.35 is the WFE cycle: The order book of ASML declined by 52 percent in one year (FY2023→FY2024) and the stock dropped 52 weeks to a low of €542 - showing material downside volatility, which fits the above-market beta profile. The sell-side consensus has a median of c. +1,400 (range 955-1700) (stronger growth and margin assumptions than our base case). Negative alpha is thus a model-optimized HOLD signal - not an inherent SELL signal..

8. MARKET NEWS, RISKS & SCENARIO ANALYSIS

Table 8.1: Key Investment Risks and Scenario Implications

Risk Factor	Severity	Analysis
China Export Restrictions	**HIGH**	Dutch controls progressively restrict DUV/EUV to China (historically ~30% of revenue). Full embargo could remove €4–6bn from FY2026E revenue, depressing intrinsic value to ~€480–500/share.
Semiconductor Cyclicality	**MEDIUM**	FY2024 order intake collapse (−52% YoY) demonstrated cycle vulnerability. A second destocking cycle - triggered by AI capex rationalisation or recession - could suppress FY2027–28 revenues.
AI Demand Sustainability	**MEDIUM**	Hyperscaler AI capex is concentrated; efficiency improvements (model compression, smaller chips)

		could reduce hardware requirements faster than consensus assumes, dampening downstream TSMC/Samsung demand.
Multiple Compression / Overvaluation Re-Rating	**HIGH**	At 40× NTM P/E (+68% vs. peers), ASML's multiple compressed from 48× (FY2021) to 29× (FY2025) historically. A re-rating to 30× (still at a 26% premium) implies ~€741 - a 35.5% decline from today.

BULL CASE → Implied: ~€900–1,100	**BEAR CASE → Implied: ~€420–500**
High-NA EUV volumes +30–40% vs. model. IBM grows at 12%+ CAGR. WACC compresses to 8% on AI re-rating. FY2030 revenue reaches €55bn. Gross margin expands to 58%. TSMC N2/A16 pull-forward of 3–4 additional EUV tools per year. China partial re-entry via licensing scheme.	Full China embargo removes €5–7bn revenue. AI capex rationalisation delays EUV orders 2 years. High-NA yields disappoint; ramp pushed to FY2029. Memory cycle freeze extends. Multiple compresses to peer average (23.75× P/E). Stock re-rates to €750–850 near-term before further downside.

9. ETHICAL CONSIDERATIONS

Market Concentration & Monopoly Power

The EUV monopoly of ASML is a type of technological gatekeeping previously unknown in modern industrial history. In contrast to differentiated competition, the customers of ASML cannot threaten substitution credibly - EUV is a mandatory component of sub-5nm manufacturing, and the power of ASML is limited by the sensitivity of the total-cost-of-ownership of the customers rather than competition. This concentration poses systemic vulnerability to the world economy at High-NA pricing of 350m+/unit. So far, ASML has been responsible in exercising its pricing power - natural moderating forces are the interest congruence with the customers (ASML thrives when chipmakers thrive). Nevertheless, with regulatory scrutiny of tech sector monopolies increasing world-wide, the market power of ASML is the principal investment virtue in the firm, and the most material long-term regulatory risk in the firm - a two-sidedness that conventional sell-side research seldom critically examines.

Export Controls & Geopolitical Weaponisation

The application of Dutch export control policy - practically at the request of US needs of national security - as an instrument to regulate the commercial operations of ASML is a fundamental structural change in the connection between commercial business and the geopolitical strategy. As an investor, the asymmetry is even more pronounced: the downside (total embargo, loss of revenue of up to 46bn+) can be estimated; the upside (relaxation of controls) is politically improbable within 35 years. More importantly, the history of relying on a privately-owned monopolist as a tool of national security would cast reasonable doubts on the

long-term optimisation of the commercial interests of ASML that are not tied to the priorities of Dutch and US governments - a risk that is systematically underweighted in consensus valuation models.

ESG Considerations

The direct environmental footprint of ASML is comparatively limited - committed to a carbon neutral make-up by 2025 in operations, net-zero supply chain by 2040. The products it makes make it possible to create energy-efficient chips, which is beneficial to decarbonisation in the world. Nevertheless, the semiconductor supply chain instantiates material ESG challenges: PFAS-based photoresists, rare earths in optical components, and an extraordinary amount of water in state-of-the-art fabs are issues that have been ascribed to the downstream ecosystem of ASML. The twofold character of the EUV-neighboring precision optics technology - which may be utilized in military precision targeting - establishes technology stewardship duties which cannot be appropriately reflected in the standard ESG rating frameworks. Investors aiming to achieve ESG-consistent portfolios ought to engage in the scenario analysis within a truly comprehensive supply-chain ESG prism, rather than just ASML direct operational indices.

10. FINAL INVESTMENT RECOMMENDATION

HOLD | PT €578 | Downside (49.8%) | DCF €551 | Comps €604 | Alpha −31.1%

My HOLD rating is the resolution to the tension between a pair of equally valid conclusions:

(1) ASML is an outstanding business - EUV monopoly, 34.6% EBIT margin, 15.1% revenue CAGR, -11bn net cash, -38.8bn backlog, best-in-class FCF conversion;

(2) at 1,150, the price incorporates a I'm not recommending SELL because: ASML fundamentals are not weakening; FY2025 performance was the best in the company history;

(3) FY2026 High-NA deliveries have momentum-driven appreciation and earnings beat risk; and FY2026 High-NA deliveries, not weakening;

(4) FY2026 deliveries are the best in the history of the company; High-NA is commer HOLD retains, on an existing basis, and recognizes that the risk-reward is not attractive to new capital at present prices. My re-evaluation would be at an entry price of between 750-850 where the valuation premium would be more justified against fundamental value

Table 10.1: Investment Thesis Summary and Valuation Conclusion

Dimension	Assessment	Key Evidence
Business Quality	★★★★★ **Exceptional**	EUV monopoly; sole supplier for sub-5nm; 40+ year technology lead
Financial Strength	★★★★★ **Exceptional**	€11bn net cash; 114.8% FCF/NI; EPS CAGR 16.3%; buybacks at ~€800

Growth Visibility	**★★★★☆ Strong**	High-NA ramp; AI capex cycle; IBM recurring; €38.8bn backlog
DCF Valuation	**⚠ Overvalued (52.1%)**	Intrinsic €551 vs €1,150 market price; WACC 9.0%, TGR 3.0%
Comps Valuation	**⚠ Overvalued (47.4%)**	Peer-implied avg €604; ASML P/E 40× vs peer avg 23.75×
CAPM Alpha Signal	**⚠ Alpha −31.1%**	Implied return −20.9% vs required 10.2%; significant risk-reward deficit
Key Risks	**◑ Moderate-High**	China export controls; WFE cycle; AI demand concentration; multiple compression
FINAL RECOMMENDATION	**HOLD**	**Exceptional business; premium valuation. Re-assess entry at €750–850.**

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Valuation Model Source: All financial projections and valuation outputs in this report are derived from the My Excel-based model (DCF, CAPM, and comparable analysis), available at: 40527763_Asset Pricing Model